



POST-QUANTUM CRYPTOGRAPHY

*Protect Today
what must still be
Secret Tomorrow*

{**e**lysium**s**ecurity}
cyber protection & response

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Classification: **PUBLIC**

This presentation focuses on the impact future advances in computing will have on the cryptographic controls we have in place today.

Specifically, what do we need to do now to ensure we are safe tomorrow.

1. CONTEXT:

WHY THE DEADLINE IS ALREADY UPON US

2. EXPOSURE:

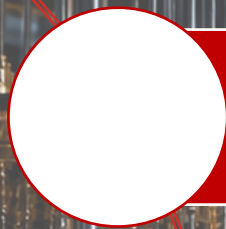
WHERE VULNERABLE CRYPTOGRAPHY HIDES

3. PLAYBOOK:

WHAT TO DO, HOW AND WHEN

4. TAKE AWAY:

WHAT GOOD LOOKS LIKE



CONTEXT



EXPOSURE



PLAYBOOK



TAKE AWAY

“QUANTUM COMPUTERS WILL BREAK ALL ENCRYPTIONS,
THERE WILL BE NO SECRETS ANYMORE,
COMMUNICATIONS AND PASSWORDS WILL BE EXPOSED”

THIS WILL BE QUANTUM DOOM’S DAY: **Q-DAY**

*Before you panic, remember Year-2000, the supposedly "end of computer world"
(a lot of work was still required though, that's the point!)*

Cyber Cryptography creates trust in untrusted places



CONFIDENTIALITY

Only Authorized parties
can read the
information



INTEGRITY

Undetected changes to
data become difficult
and noisy



AUTHENTICATION

A person, device or
service can prove its
identity or origin



NON-REPUDIATION

Digital signatures can
support evidence that
an action occurred

There are three cryptographic tools families, each used for different purposes

SYMMETRIC

(PRIVATE KEY ALGO)

One shared Secret key

Fast protection for bulk data
Ex: AES, DES, 3DES, BUGS, etc

ASYMMETRIC

(PUBLIC KEY ALGO)

Public key + Private key

Establish a secret or sign
Ex: RSA, ECC, etc.

HASH

(HASHING ALGO)

No decryption key, one way

A fixed-length fingerprint
Ex: SHA-3, SHA-2, MD5, etc.

For example, a secure web session:

1. Authenticates and agrees keys (ASYMMETRIC)
2. Encrypts the traffic (SYMMETRIC)
3. Hashes data for integrity (HASH)

CLASSICAL BIT

0 or 1

A stable state that can be copied and read directly.

QUANTUM BIT - QUBIT

$$\alpha|0\rangle + \beta|1\rangle$$

SUPERPOSITION

Gates change probability amplitudes.

INTERFERENCE

The algorithm amplifies useful outcomes.

A QUBIT carries amplitudes and measurement returns a bit (still a classical 0 or 1)

Quantum tunneling Entanglement qubit state Measurement Superposition
Wave function Bloch sphere Interference
Probability Quantum supremacy Quantum Gates
Decoherence Quantum circuit Quantum sensing

BITS power everyday computing; QUBITS can accelerate specific complex problems.

Because QUBITS can work with multiple possibilities at once

Quantum computing is not a universal skeleton key.
The effect depends on the algorithm being attacked.

SHOR'S ALGORITHM

PUBLIC-KEY

Efficiently solves the factoring and discrete-logarithm problems used by RSA, Diffie–Hellman and elliptic-curve cryptography.

IMPACT: replace mathematical assumptions

GROVER'S ALGORITHM

PRIVATE-KEY & HASH

Provides a quadratic speed-up for brute-force search. Symmetric encryption and hashes lose security margin, but are not affected like RSA or ECC.

IMPACT: review key and hash sizes

QUANTUM COMPUTING

Computers that exploit superposition, interference and entanglement.
They create the future cryptographic threat.

CRQC / Q-DAY

Cryptographically Relevant Quantum Computer (CRQC): a sufficiently capable machine.
Q-Day: the point when that capability can attack deployed public-key cryptography.

POST-QUANTUM CRYPTOGRAPHY (PQC)

New mathematical algorithms that run on ordinary computers and aim to resist classical and quantum attacks.

QUANTUM CRYPTOGRAPHY / QKD

Quantum signals, often photons, used to distribute keys and detect interception.
Requires a separate network architecture.

**This talk is about migrating ordinary IT systems to PQC
and not installing a quantum communications network!**

KEY

A secret value that controls a cryptographic operation.

DIGITAL SIGNATURE

Uses a private key to support origin and integrity checks, not confidentiality.

HYBRID

Combines classical and post-quantum components during transition.

KEM

A key-encapsulation mechanism establishes a shared secret over an untrusted network.

CERTIFICATE / PKI

Binds a public key to an identity through a chain of trust.

CRYPTO AGILITY

The capability to replace cryptography without breaking security or operations.

NUMBER OF QUBITS REQUIRED TO BREAK RSA-2048 BIT INTEGERS

2019

20 Million

Noisy QUBITS

Time required about 8 hours

Gidney–Ekerå estimate paper
<https://arxiv.org/abs/1905.09749>20x fewer QUBITS required
in 6 years

2025

<1 Million

Noisy QUBITS

Time required about 1 week

Gidney estimate paper
<https://arxiv.org/abs/2505.15917>

Not one of those reductions came from a bigger chip.
They came from better mathematics.

156

PROGRAMMABLE QUIBITS

Noisy & error prone

IBM Geron processes available
through a public cloud

70

LOGICAL QUBITS

Reliable (error correction)

temporary circuit sampling test
by IBM in July 2026

21

LARGEST FACTORED NUMBER

Using Shor Algorithm

on a quantum computer

15 in 2001 (IBM)

21 in 2012 (Photonics Paper)

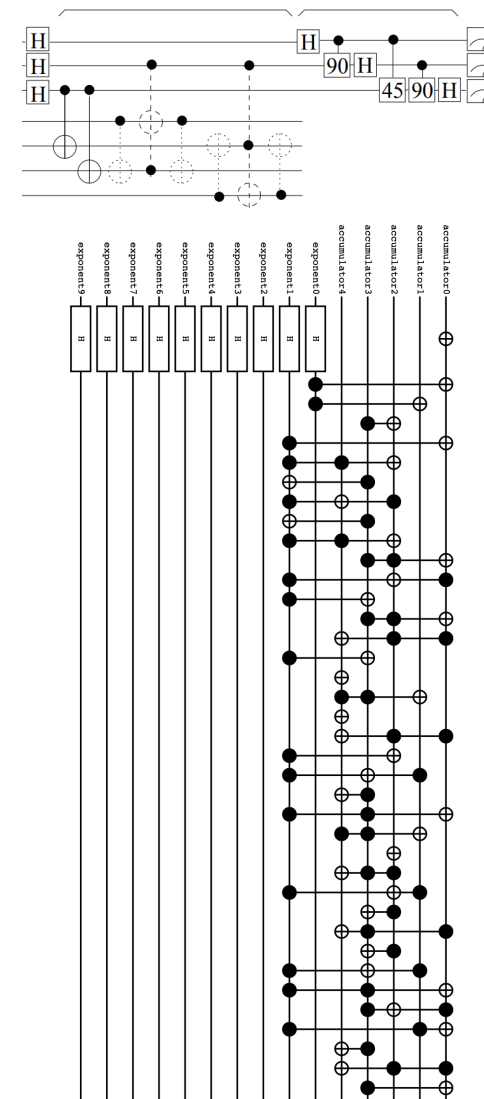
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Publicly demonstrated
Cryptographically Relevant
Quantum Computers

3

Finalized NIST PQC standards
ready for implementation

NOW

The recommended time to
inventory and planNo public CRQC exists.
The migration case still standsOPENQASM2 version of the circuit
Source: <https://algassert.com/post/2500>



Your SSH client already tells you this is happening

```
*** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.
```

OpenSSH 10.1 – released October 2025, warning on by default.

**SECRECY SHELF LIFE**

How many years this data must stay confidential
No control (law, contracts, etc) but can measure

**MIGRATION TIME**

How many years to actually replace the cryptography
Full control and ownership

**TIME TO A CAPABLE MACHINE**

Years until a (CRQC) Cryptographically Relevant Quantum Computer exists
No control

$$\text{X} + \text{Y} = \text{Z}$$

EXAMPLE: A HOSPITAL

X Patient records: **10-20 years** of required confidentiality depending on the country

Y Realistic migration of a hospital estate: **5-10 years**

Z Expert view: CRQC will exist within **10 years**

$$10 + 5 = 15$$
$$15 > 10$$

Use existing standards, supported protocols and validated products

ML-KEM

(FIPS 203)

KEY ESTABLISHMENT

Establishes shared secrets; it
does not sign.
RSA replacement.

ML-DSA

(FIPS 204)

DIGITAL SIGNATURES

General-purpose post-quantum
signatures.
Fast and versatile.

SLH-DSA

(FIPS 205)

DIGITAL SIGNATURES

Hash-based alternative with
larger signatures.
Heavy and Fall back



→ Linked to NIST SP 800-227 (general guide for KEM)

28 JULY 2026

HAWK

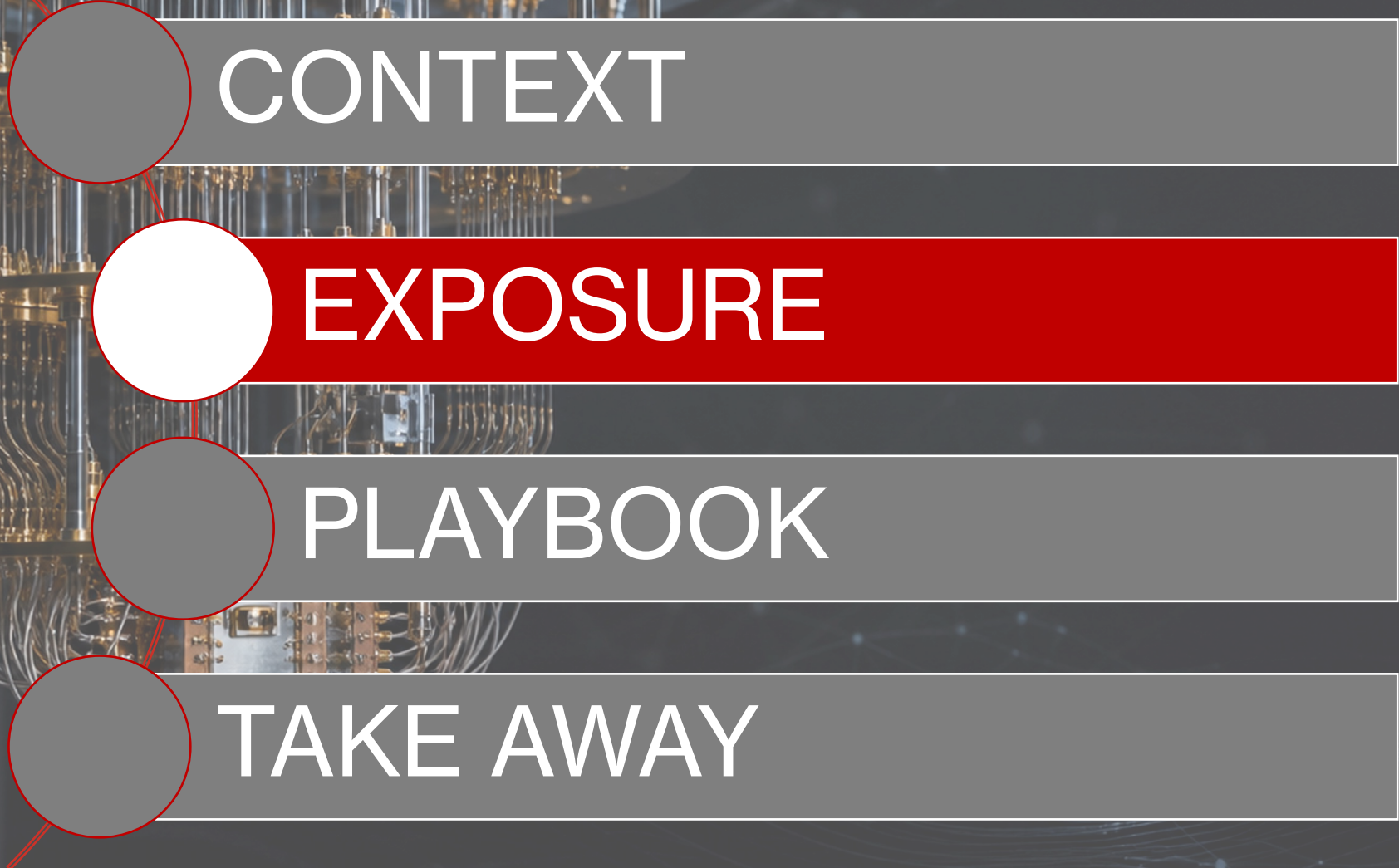
A signature candidate was withdrawn after an AI-assisted mathematical vulnerability was disclosed.



WHAT THIS MEANS

- The review process worked: a candidate was removed before standardization.
- NIST says the finalized standards were not affected.
- Your design must support change without a new multi-year project.

THE DURABLE CONTROL IS CRYPTO AGILITY.



CONTEXT

EXPOSURE

PLAYBOOK

TAKE AWAY

BROKEN BY SHOR'S ALGORITHM***REPLACE***

- RSA - every key size, encryption and signing
- Finite-field Diffie–Hellman / DSA
- ECC: ECDH, ECDSA, EdDSA

ONLY WEAKEN BY GROVER***KEEP & RESIZE***

- Symmetric Encryption - AES-256
- Hash Functions - SHA-384/512
- Use policy-approved parameters

SO WHAT DOES THAT MEAN IN YOUR ESTATE?

TLS / HTTPS

VPN & IPsec tunnels

SSH and file transfer

Wi-Fi enterprise auth

Certificates & internal PKI

Code and firmware signing

Smart cards, Bank payment cards, passports

HSMs and key ceremonies

Secure email (S/MIME, PGP)

Database & backup key wrapping

Blockchain wallets and ledgers

IoT, OT and metering devices

'Encryption' will not break soon. Key exchange and signatures might do.
Your AES-256 backup is fine. The key that unlocked it and travelled over RSA, is not!

ALREADY DONE FOR YOU, FREE

- **Chrome, Edge and Firefox:** hybrid X25519 + ML-KEM by default
- **Apple iOS 26** (Sept 2025): quantum-secure TLS on by default
- **OpenSSH 10.0** (Apr 2025): post-quantum key exchange default
- **Signal:** post-quantum ratchet (SPQR) shipped October 2025
- **Cloudflare, AWS, Google and Meta edges:** hybrid TLS enabled
- **Result:** 1/3 of human web traffic in Jan 2025 -> 2/3 today

NOT DONE FOR YOU

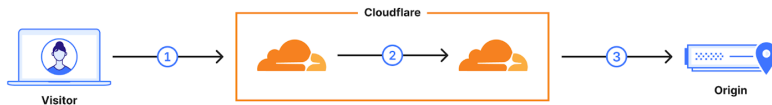
- Site-to-site VPNs, firewalls and SD-WAN
- HSMs, key ceremonies, internal CA and certificate estate
- Code signing, firmware signing, software supply chain
- SFTP / MFT feeds to banks, regulators and partners
- Database, backup and archive key wrapping
- ATMs, POS, smart cards, badges, smart meters, etc.

TEST IT YOURSELF

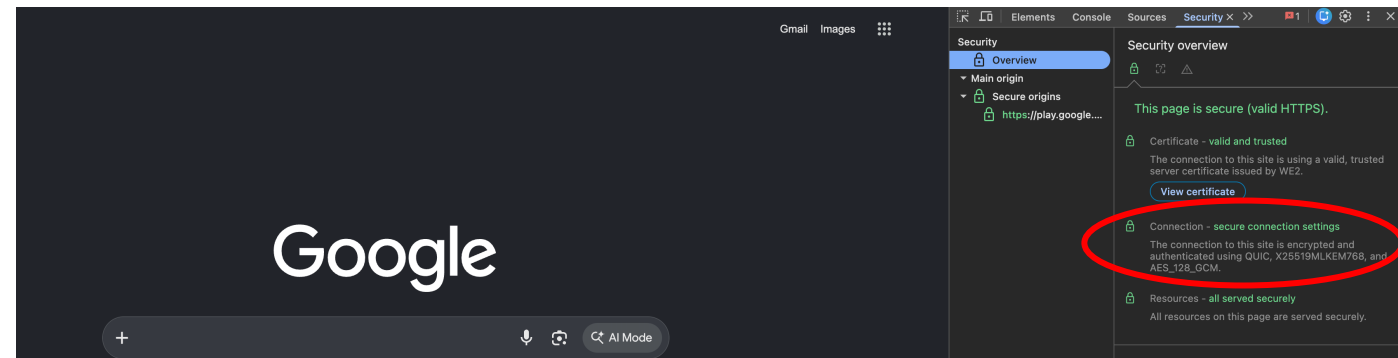
Your Browser: <https://pq.cloudflareresearch.com/>

The website you visit

Cloudflare Research: Post-Quantum Key Agreement



You are using *X25519MLKEM768* which is **post-quantum secure**.



CONFIDENTIALITY***DEADLINE: TODAY***

- Attackable retroactively through collected traffic or archives.
- Prioritise long-lived data and exposed transport paths.
- Hybrid key exchange may reduce risk where supported and justified.

AUTHENTICITY***DEADLINE: TRUST CYCLE***

- A future forgery matters while the verifier still trusts the classical key.
- Roots, HSMs, firmware, devices and smart cards create long lead times.
- Plan signature migration around renewal and technology-refresh cycles.

Confidentiality is a race against a machine.
Authenticity is a race against your own renewal cycle.



Jurisdiction-specific policy signals – not a universal compliance timeline.
Common message: Inventory -> Priorities -> (Supplier) plans -> Multi-year execution

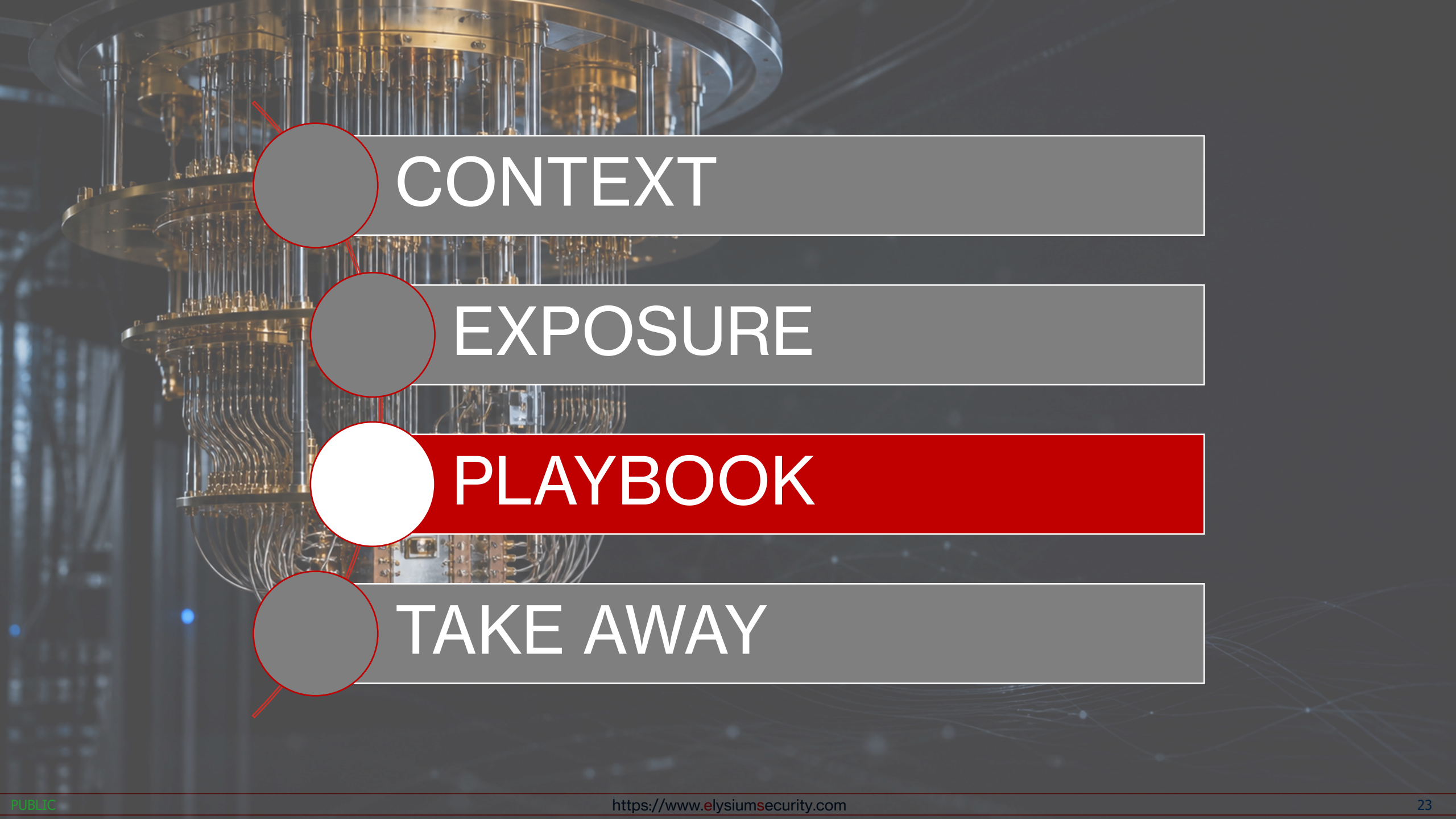
THE RULES THAT ALREADY MOVED

- Bank of Mauritius Guideline on Cyber and Technology Risk Management (effective 29 May 2023), s.28(i): cryptographic controls must follow "*recognised standards and processes... algorithm, key length, and key generation*"
- Since 13 August 2024 the recognised standard includes ML-KEM and ML-DSA.
- Data Protection Act 2017: "*appropriate technical measures*" are judged against the risk. For harvested data, that judgement lands in the year it is finally decrypted.

HOW THE DEADLINE WILL ARRIVE HERE

- Parents company in Europe, India and South Africa exporting policy
- International banks passing their own 2030 obligations down the chain
- EU clients under NIS2 and DORA assessing you as a third-party supplier
- Card schemes and payment processors updating technical requirements
- Cloud and SaaS providers changing defaults and deprecating old suites

The deadline may not first arrive from a regulator.
It will likely arrive in a supplier questionnaire.



CONTEXT

EXPOSURE

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TAKE AWAY

1

What cryptography do you actually run?

Algorithms, key lengths, where the keys live, and which vendor owns each one. Not a diagram, a list.

→ *Most important point*

2

What is your longest-lived secret, and when does it stop mattering?

One line per data type. The answer decides your priority order, not the age of the system.

3

Who owns this?

Name a person, not a department. A migration without an owner is a slide deck.

→ *usually owns by nobody today*

4

What have your vendors committed to, in writing, with a date?

Verbal reassurance from an account manager is not a roadmap. Ask for the release and the version.

5

Could you change an algorithm without running a project?

If changing a cipher requires a business case, you do not have a migration problem: you have an agility problem.

NO/LOW
COST

YEARS
TO
ACHIEVE

NETWORK EDGE

TLS versions and cipher suites, load balancers, reverse proxies, VPN and IPsec, SD-WAN, Wi-Fi enterprise auth.

CERTIFICATES & PKI

Internal CA, root and issuing key algorithms, anything valid beyond 2030, certificate discovery across the estate.

CODE & PIPELINES

Hardcoded algorithms and key sizes, crypto libraries and pinned versions, signing and release pipelines.

KEYS, HSMs & KMS

Key wrapping, escrow, ceremonies, cloud KMS defaults, which HSM firmware supports which algorithms.

VENDORS & SAAS

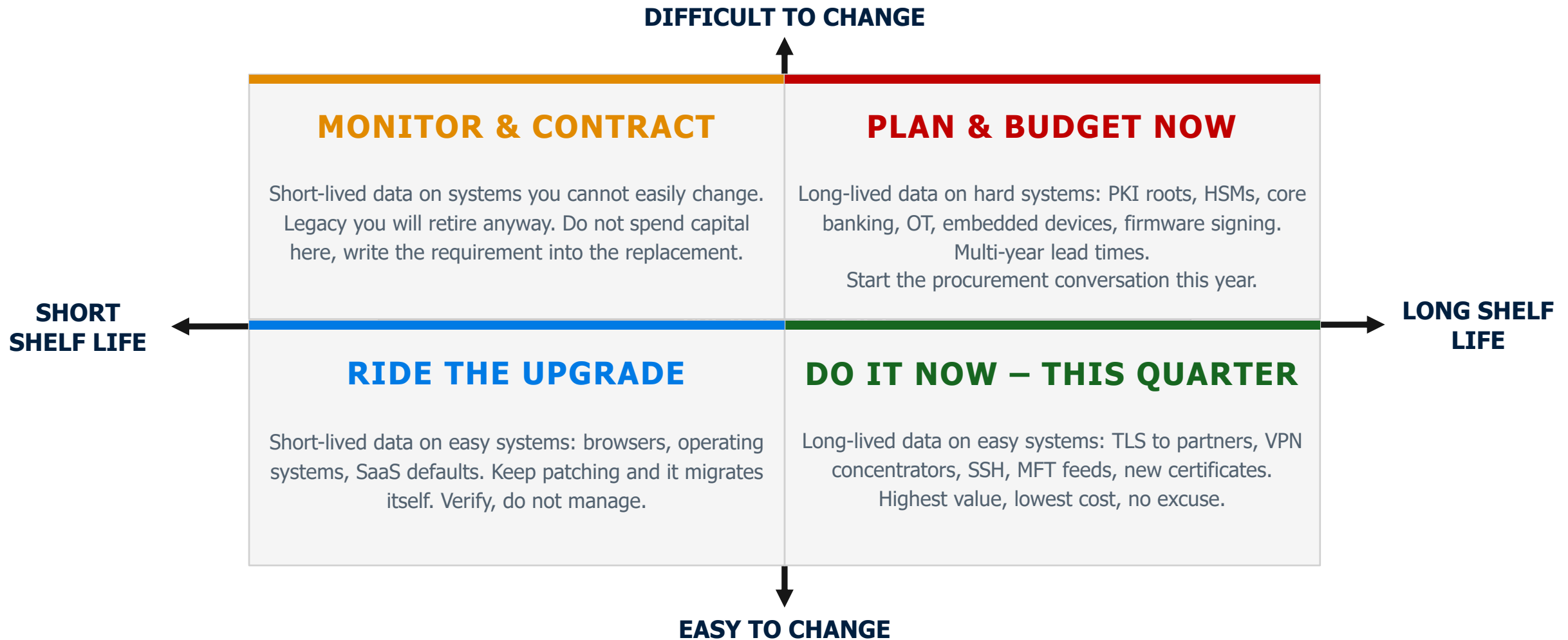
Every supplier's stated algorithms, roadmap dates and version numbers: captured in the contract file, not in an email.

OT, IOT & CARDS

Firmware signing, ATMs, POS terminals, smart cards, badges, meters, medical devices, anything with a 10-year life.

You cannot migrate what you cannot list.

A simple spreadsheet done within a month will save you future expenses



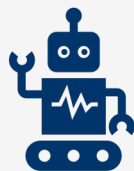
Start bottom-right. It is the cheapest quadrant, and it closes the harvesting window first.

- 1 Enable hybrid post-quantum key exchange at your CDN, edge and load balancers, usually a single setting.
- 2 Upgrade to OpenSSH 10.x and stop dismissing the "store now, decrypt later" warning.
- 3 Push browser and operating system updates, a large share of your exposure closes by itself.
- 4 Ask your VPN, firewall and MFT vendors for their post-quantum release. Several already ship it and nobody enabled it.
- 5 Move symmetric cryptography to AES-256 and SHA-384 wherever it is a configuration option.
- 6 **Stop issuing long-lived RSA-2048 certificates, and stop signing roots that expire after 2030 without a plan.**
- 7 Add "post-quantum roadmap, with version numbers and dates" to your standard supplier questionnaire.

Doing these seven closes the harvesting window on most of your traffic before your budget cycle even opens.

**ABSTRACT**

No hardcoded algorithm names or key sizes in business logic.

**AUTOMATE**

Issue, rotate and revoke keys and certificates without tickets.

**OBSERVE**

Know what is negotiated, where it failed and what remains exposed.

**CHANGE AGAIN**

Maintain inventory, rollback and supplier evidence as standards evolve.

PUBLIC TLS CERTIFICATE
MAXIMUM VALIDITY
EXAMPLE



There will be more than one migration. Build the ability to change, not a single change.

SUPPORTED NOW

Which NIST algorithms, parameters and protocols are in which product version today?

ROADMAP

Separate dates for hybrid key exchange, post-quantum signatures and validation.

CHANGEABILITY

Can the algorithm change without replacing hardware or breaking the service contract?

DATA ALREADY HELD

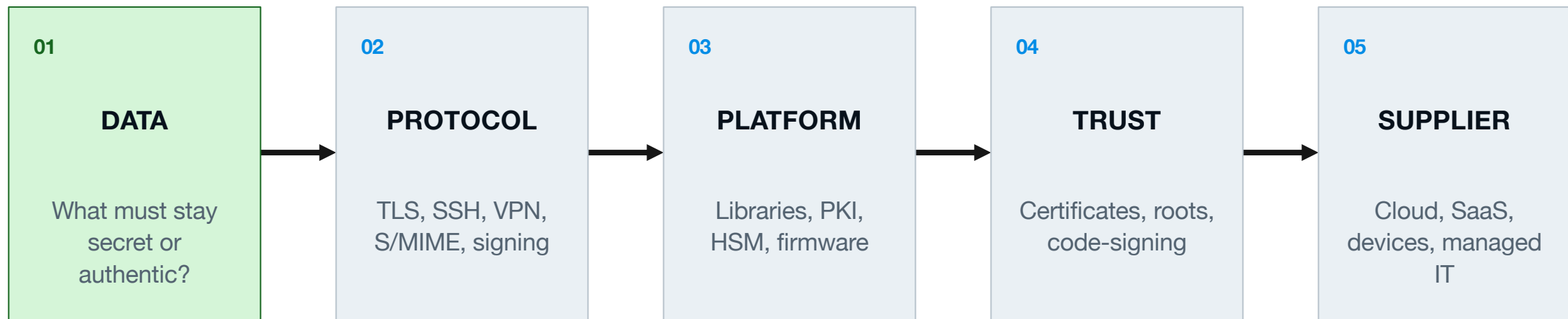
How are our backups, archives, keys and replicas protected through the transition?

EVIDENCE

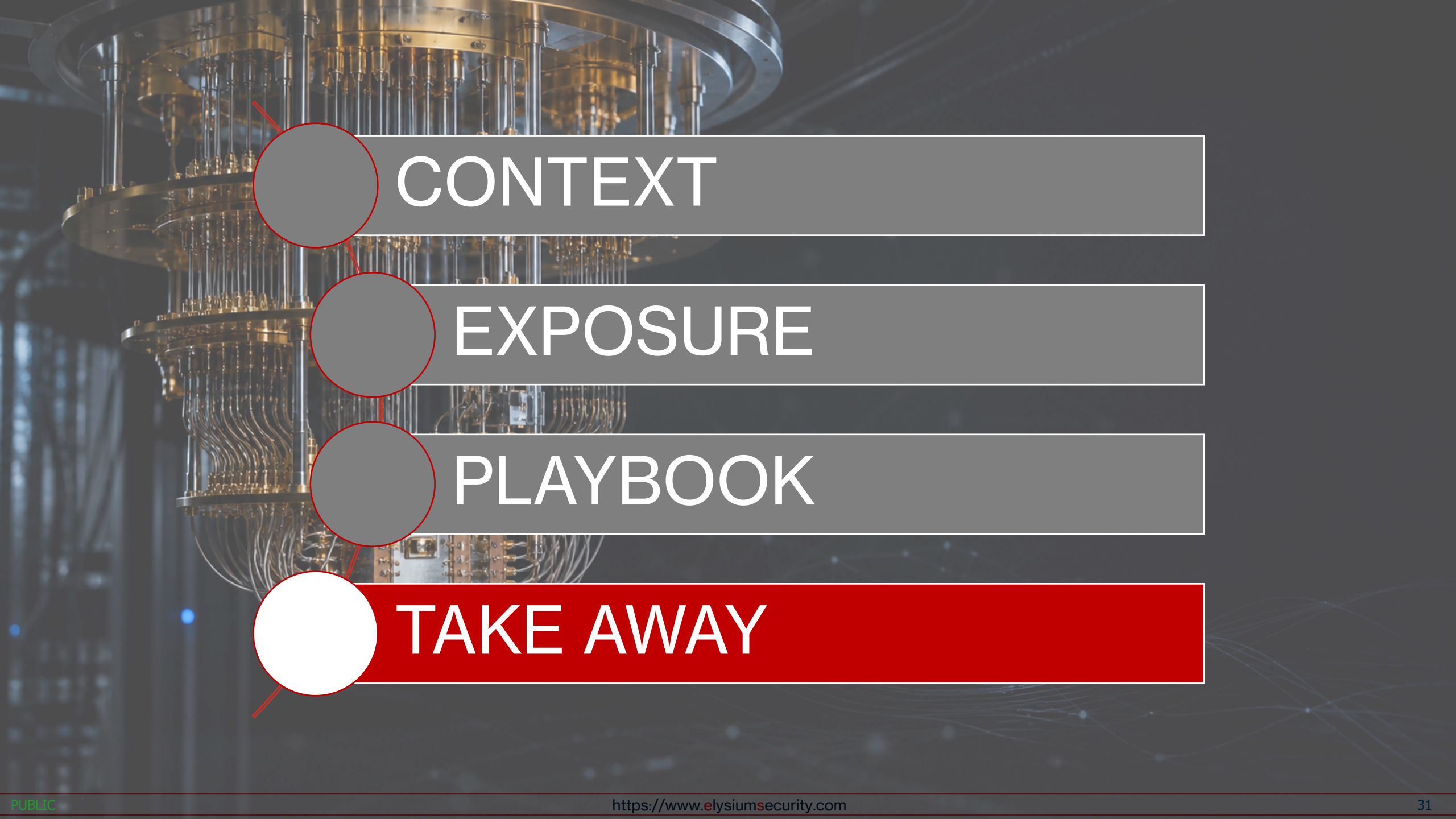
What testing, interoperability, rollback and certification evidence will you provide?

Do not buy 'quantum-safe.'

Buy a supported version, a date, a test plan and a rollback path.



Change any one link without testing the others and the service may still be vulnerable, or stop working



CONTEXT

EXPOSURE

PLAYBOOK

TAKE AWAY

OWN

A named risk owner and decision forum.

KNOW

A prioritised and maintained inventory linked to business services

CHANGE

Tested migration, automation and rollback paths.

PROVE

Supplier commitments, metrics and evidence.

Four repeatable capabilities

Three are **governance** capabilities. Only one is primarily **cryptographic**.

THE OBJECTIVE IS NOT TO BE QUANTUM-SAFE BY FRIDAY

Know what you run. Protect what must still be secret in 2035. Be able to change again.

THIS MONTH**Name the owner.**

One person, accountable, on the risk register. Not a department, not a committee.

NEXT MONTH**Start the inventory.**

External TLS scan, certificate discovery, and a questionnaire to your ten most important suppliers.

THIS QUARTER**Turn on the free wins.**

Edge and load balancer hybrid TLS, OpenSSH 10.x, OS and browser updates, AES-256 where it is a setting.

ONE MEETING**Write down your three longest-lived secrets.**

And the year each one stops mattering. That single page sets every priority that follows.

FROM NOW ON**Put a product version and dates in every contract.**

Post-quantum roadmap, version number and date: in every renewal, RFP and third-party assessment.

Nobody can finish this migration in 2026.
Everybody can start it in 2026.

STANDARDS AND TECHNICAL

- NIST FIPS 203 / 204 / 205 (13 Aug 2024) — ML-KEM, ML-DSA, SLH-DSA
- NIST IR 8547 (draft) — transition plan: deprecate 2030, disallow 2035
- NIST IR 8545 — HQC selection, March 2025
- NIST SP 800-227 — key-encapsulation mechanisms, Sept 2025
- Gidney, arXiv:2505.15917 — RSA-2048 with under a million qubits
- Cain et al., arXiv:2603.28627 — Shor with $\sim 10,000$ atomic qubits

POLICY, REGULATION AND PRACTICE

- US Executive Order 14412 (22 June 2026)
- EU Coordinated Implementation Roadmap for PQC (June 2025)
- UK NCSC — timelines for migration to post-quantum cryptography
- NSA CNSA 2.0 — national security systems timeline to 2033
- Global Risk Institute — Quantum Threat Timeline Report 2025
- Federal Reserve FEDS 2025-093 — harvest now, decrypt later
- Cloud Security Alliance — AI infrastructure and HNDL (May 2026)
- Bank of Mauritius — Guideline on Cyber and Technology Risk Management

THANK YOU!

Any Questions?



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